ANOVA One Way

Table of contents

```r
data("ToothGrowth")
```

```r
library(car)
```

```
Loading required package: carData
```

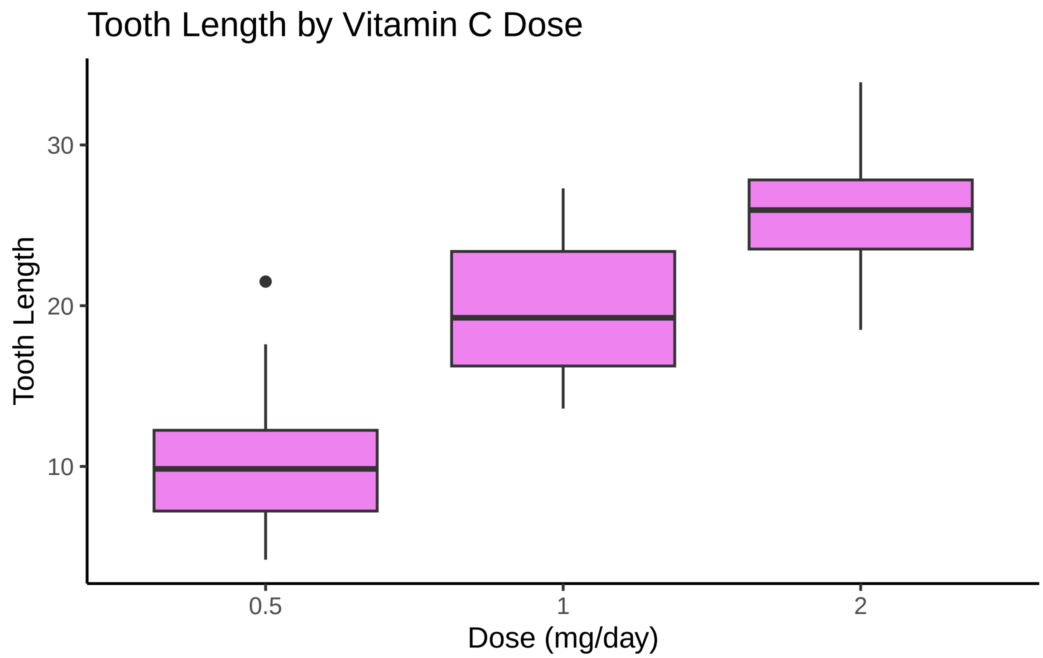
```r
# Convert dose to a factor
ToothGrowth$dose <- as.factor(ToothGrowth$dose)

# ANOVA One-Way
anova_result <- aov(len ~ dose, data = ToothGrowth)
summary(anova_result)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
dose         2   2426    1213   67.42 9.53e-16 ***
Residuals   57   1026      18                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Assumption check
leveneTest(len ~ dose, data = ToothGrowth)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.6457 0.5281
      57               
```

```r
# Box-plot
library(ggplot2)
ggplot(ToothGrowth, aes(x = dose, y = len)) +
  geom_boxplot(fill = "violet") +
  labs(title = "Tooth Length by Vitamin C Dose",
    x = "Dose (mg/day)", y = "Tooth Length") +
    theme_classic()
```




```r
#Post-Hoc Test
TukeyHSD(anova_result)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = len ~ dose, data = ToothGrowth)

$dose
        diff       lwr       upr    p adj
1-0.5  9.130  5.901805 12.358195 0.00e+00
2-0.5 15.495 12.266805 18.723195 0.00e+00
2-1    6.365  3.136805  9.593195 4.25e-05
```

Null Hypothesis (H0): There is no difference in the averages among the three different Vitamin C dose groups (0.5 mg, 1 mg, and 2 mg).

Alternative Hypothesis (H1): There is a difference (at least two) in the average tooth length among the dose groups.

In the ANOVA one-way test, I checked to see if different amounts of Vitamin C affected how much tooth grew for guinea pigs. The test showed a statistically significant difference between the groups, meaning that at the very least (potentially), one of the dose levels led to a different amount of tooth growth.

The boxplot also shows a visible difference in the median tooth length between the three dose groups.

Therefore, I reject the null hypothesis. Vitamin C does affect guinea pig's tooth growth.

At the post-hoc test, I found that all three comparisons were statistically significant, with higher doses having greater tooth growth.

**Only Vitamin C was looked upon, but not Orange Juice from the dataset.